

1. Administrative & Study Identification

Full Study Title: A Clinical Study of MK-2214 in People With Early Alzheimer's Disease (MK-2214-004) **Short Title/Acronym:** MK-2214-004 **Protocol Number:** MK-2214-004 **NCT Number:** NCT07033494 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** MK-2214 (Anti-tau monoclonal antibody) **Phase of Development:** Phase 2 **Medical Monitor:** Lawren VandeVrede, MD, PhD (Study Director)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of MK-2214 in slowing the spread of tau protein in the brain compared to placebo in participants with early Alzheimer's disease (AD). **Secondary Objectives:** To evaluate the overall safety, tolerability, and pharmacokinetics of MK-2214.

2.2 Background & Rationale To address the progression of early Alzheimer's disease and cognitive decline. Tau is a protein that abnormally accumulates in Alzheimer's disease and damages brain cells, directly contributing to disease progression. This study investigates whether MK-2214—an investigational novel antibody targeting phosphorylated serine 413 (pS413) tau—can slow the spread of tau pathology and potentially slow the clinical progression of the disease.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 340 participants **Number of Sites:** Multiple clinical sites across the United States (e.g., UCSF Memory and Aging Center, Yale University ADRU, etc.) **Estimated Study Start:** 16-Jul-2025 **Estimated Primary Completion:** 30-Apr-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 50 to 85 years. 2. Confirmed diagnosis of mild cognitive impairment (MCI) or mild dementia due to Alzheimer's disease (AD). 3. Must have a designated study partner who can fulfill the requirements of assisting with study

procedures. 4. If receiving symptomatic AD therapy, the dosing regimen must be stable for at least 3 months prior to screening.

4.2 Exclusion Criteria 1. History of stroke, cerebrovascular disease, or a clinically relevant central nervous system disease other than AD. 2. History of seizures or epilepsy within 5 years before screening, or structural brain disease. 3. Any immunological disease that is not adequately controlled, or which requires treatment with biologics and/or immunosuppressants during the study. 4. Unwilling or unable to undergo computed tomography (CT), positron emission tomography (PET), or magnetic resonance imaging (MRI) scans.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered every 4 weeks (q4w). **Route of Administration:** Intravenous (IV) Infusion. **Duration of Treatment:** Up to 23–26 months of participation.

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of standard symptomatic Alzheimer's disease medications. **Prohibited:** Concomitant treatment with other biologics and/or immunosuppressants.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -X to 0	Informed Consent, Neurological and physical exams, Baseline MRI/PET scans, Lumbar puncture
Baseline	Day 0	Randomization, First IV Infusion
Interim	Every 4 Weeks	IV Infusions, Safety Labs, Cognitive testing, Neuropsychiatric assessments, EKGs, Telephone check-ins
End of Study	Month 23-26	Final Efficacy Assessment, Follow-up PET scans, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from baseline in tau PET Standardized Uptake Value ratio (SUVR). **Measurement Method:** Positron Emission Tomography (PET) imaging (SUV in the region of interest divided by SUV in a reference region, such as the cerebellum).

7.2 Secondary & Exploratory Endpoints 1. Safety and tolerability profiles (Incidence of Adverse Events/Serious Adverse Events). 2. Changes in clinical, functional, and cognitive status over the study duration.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via regular telephone check-ins, scheduled physical examinations, EKGs, and laboratory tests. **Serious Adverse Events (SAEs):** Standard reporting timeline, typically within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent External Data Monitoring Committee to oversee patient safety and trial integrity.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy outcomes; Safety Set for AE/SAE evaluations. **Sample Size Justification:** $N = 340$, powered to detect statistically significant differences in tau PET SUVR spread between the MK-2214 and placebo arms. **Handling of Missing Data:** Multiple Imputation (MI) and Mixed-Model Repeated Measures (MMRM) for continuous imaging and cognitive endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard risk-based onsite and remote monitoring visits. **Audit Trail:** Robust electronic data capture (EDC) with full eCRF locking mechanisms and rigorous data clarification tracking.

11. Study Identifiers & Governance

Primary ID: MK-2214-004 **Registry Number:** NCT07033494 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Clinical Study of MK-2214 in People With Early Alzheimer's Disease (MK-2214-004) **Official Regulatory Title:** Phase 2 Randomized, Placebo-Controlled, Double-Blind Study to Evaluate the Efficacy and Safety of MK-2214 in Participants with Early Alzheimer's Disease

12. Clinical Framework (Alzheimer's Specific)

Primary Condition: Early Alzheimer's Disease **Diagnosis Threshold:** Mild cognitive impairment (MCI) or mild dementia due to AD **Medical Methodology:** Investigational monoclonal antibody therapy targeting pS413 tau **Optimization Target:** Slowing the pathological spread of tau protein and preserving cognitive function

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for Early AD trials including advanced imaging and biomarker collection **Education Protocol (HCP):** Site initiation and rigorous certification for standardized cognitive/neuropsychiatric raters and imaging capture protocols **Education Protocol (Patient):** Informed consent management requiring joint participation and training of the designated study partner **Quality Audit Mechanism:** Centralized independent review and reading of all MRI and PET scans

14. Observation & Follow-Up Schedule

Total Duration: Up to 23-26 months **Onsite Visit Cadence:** Every 4 weeks for IV infusions and evaluations **Remote Monitoring Frequency:** Intermittent regular telephone check-ins **Checklist Review Interval:** Every 4 weeks (Q4W) matching the infusion schedule

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				